

BEYA, Russian Federation

WMO: 299620

Lat: 53.050N

Lon: 90.917E

Elev: 468

StdP: 95.82

Time Zone: 7.00 (E07)

Period: 86-01

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.5	-27.2	-32.9	0.2	-30.6	-29.8	0.3	-27.1	11.9	-10.1	9.3	-11.1	0.4	180	0.586

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.5	28.4	17.8	26.7	17.5	25.1	16.7	19.5	26.0	18.6	24.8	17.6	23.5	2.3	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.1	12.9	22.8	16.1	12.1	21.7	15.2	11.4	20.9	57.9	26.0	54.8	25.1	51.6	23.5	23.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.7	7.3	6.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.5	-14.9	-11.4	-4.8	3.8	11.5	16.1	18.2	16.4	8.8	2.6	-7.5	-10.3
	DBStd	12.62	6.87	6.21	6.84	4.69	4.89	3.99	3.04	3.27	4.07	4.92	7.48	6.27
	HDD10.0	3525	772	601	460	190	42	4	0	1	69	233	525	629
	HDD18.3	5875	1031	834	718	434	217	89	38	78	286	488	775	887
	CDD10.0	771	0	0	0	6	88	186	256	198	34	3	0	0
	CDD18.3	78	0	0	0	0	4	21	35	17	1	0	0	0
	CDH23.3	746	0	0	0	0	85	194	287	177	3	0	0	0
	CDH26.7	151	0	0	0	0	17	36	59	39	0	0	0	0
Wind	WSAvg	2.2	1.7	2.1	2.2	2.9	2.6	2.2	1.7	1.8	2.1	2.4	2.5	2.3
Precipitation	PrecAvg	365	8	6	9	20	38	68	73	65	36	18	14	9
	PrecMax	559	18	14	34	47	109	138	113	115	80	59	35	18
	PrecMin	257	1	1	2	3	11	27	44	27	12	4	4	2
	PrecStd	63	5	3	7	13	21	27	18	27	16	11	9	4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.7	2.7	14.8	20.7	28.8	29.9	30.9	30.3	23.1	17.2	8.7	4.5
		MCWB	-1.9	-0.6	7.4	9.8	15.5	17.5	19.6	18.8	14.2	9.2	3.3	1.7
	2%	DB	-1.3	0.7	9.3	17.9	25.9	27.6	28.1	27.5	20.3	14.9	5.5	1.7
		MCWB	-3.4	-2.0	4.2	8.7	14.0	17.0	19.0	17.9	12.9	8.1	1.8	-0.8
	5%	DB	-3.5	-0.9	6.0	15.5	23.5	25.6	26.5	25.2	18.5	12.8	4.0	-0.1
		MCWB	-5.4	-3.3	1.9	7.7	13.1	16.7	18.3	17.2	12.2	7.4	1.0	-2.4
	10%	DB	-5.2	-2.4	3.5	12.7	21.1	23.8	24.8	23.2	16.1	10.0	2.4	-1.7
		MCWB	-6.9	-4.5	0.0	6.5	12.1	16.0	17.6	16.5	10.9	5.9	-0.2	-3.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.6	-0.4	7.6	11.2	16.9	19.9	20.9	20.3	16.0	11.5	3.9	2.1
		MCDB	0.5	2.1	14.2	19.2	26.5	26.2	27.2	27.2	21.1	15.7	7.7	4.3
	2%	WB	-3.4	-1.8	4.6	9.3	15.0	18.4	19.7	19.0	14.1	9.2	2.3	-0.7
		MCDB	-1.4	0.4	9.2	16.3	23.4	25.2	26.5	25.5	18.9	13.5	5.1	1.4
	5%	WB	-5.2	-3.1	2.3	8.0	13.9	17.3	19.0	17.9	12.6	7.5	1.1	-2.2
		MCDB	-3.6	-0.9	5.8	14.4	21.8	24.1	25.4	24.2	17.2	12.0	4.0	-0.2
	10%	WB	-6.9	-4.6	0.2	6.7	12.7	16.4	18.1	16.8	11.3	6.0	-0.2	-3.8
		MCDB	-5.3	-2.4	3.2	12.6	19.9	22.9	23.7	22.2	15.5	10.0	2.4	-1.8
Mean Daily Temperature Range	5% DB	MDBR	8.5	9.1	9.6	10.7	11.6	10.0	9.5	10.4	9.3	8.3	7.7	7.7
		MCDBR	10.5	10.4	11.6	15.6	16.0	13.3	13.1	14.3	13.3	12.4	9.0	9.2
		MCWBR	8.9	8.6	7.4	8.7	7.1	5.8	5.6	6.8	6.9	7.0	6.5	7.9
	5% WB	MCDBR	10.1	10.1	11.2	13.8	13.2	11.7	11.9	12.4	11.5	11.7	8.4	9.4
		MCWBR	8.9	8.5	7.4	8.0	6.2	5.9	5.5	6.3	6.7	7.3	6.1	8.3
Clear-Sky Solar Irradiance	taub		0.280	0.316	0.365	0.409	0.413	0.400	0.408	0.394	0.357	0.345	0.319	0.285
	taud		2.074	2.023	2.016	2.135	2.180	2.319	2.320	2.331	2.372	2.272	2.125	2.030
	Ebn at Noon		669	757	798	817	836	852	836	827	816	730	609	568
	Edh at Noon		73	107	134	136	137	121	119	111	94	84	70	64
All-Sky Solar Radiation	RadAvg		1.09	2.13	3.55	4.53	5.30	6.04	5.50	4.64	3.35	1.91	0.99	0.72
	RadStd		0.08	0.14	0.26	0.28	0.46	0.51	0.37	0.36	0.30	0.18	0.11	0.07

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page